

CFTF: Controllable Fine-grained Text2Face and Its Human-in-the-loop Suspect Portraits Application

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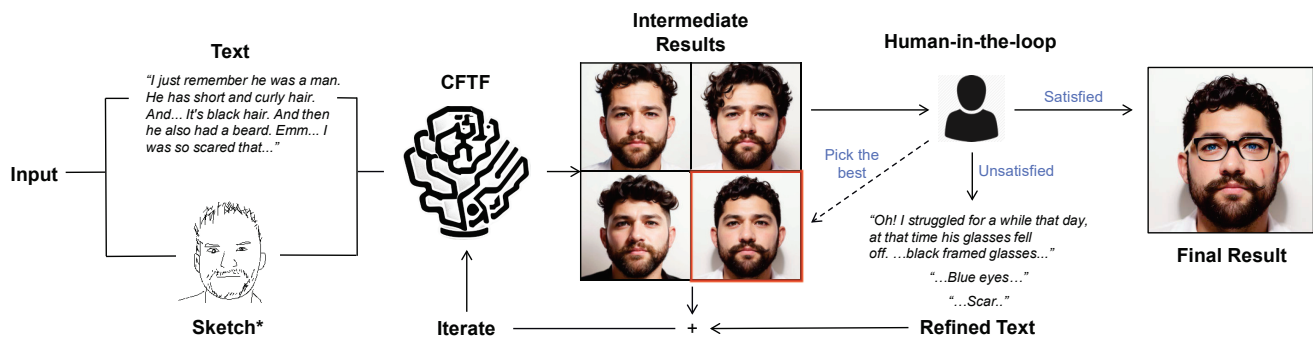


Figure 1: The system workflow and the core contributions of our work: 1) CFTF: Emphasizes controllable fine-grained Text2Face; 2) HITL: A suspect portrait application that integrates user feedback into the face generation process, including user's initial input (Text + Sketch*), user's selection of intermediate results, and more detailed text descriptions during the iteration process.

ABSTRACT

The traditional controllable face generation refers to the controllability of coarse-grained ranges such as facial features, expression postures, or viewing angles, but specific application scenarios require finer-grained control. This paper proposes a fine-grained and controllable face generation technology, CFTF. CFTF allows users to participate deeply in the face generation process through multiple rounds of language feedback. It not only enables control over coarse-grained features such as gender and viewing angle, but also provides flexible control over details such as hair color, accessories, and iris color. We apply CFTF to the suspect portrait scene, and perform multiple rounds of human-computer interaction based on the eyewitness's painting sketch of the suspect and descriptions of their facial features, realizing the "Human-in-the-loop" [9] collaborative portrait drawing.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision**; • **Human-centered computing** → **Interaction techniques**.

KEYWORDS

AIGC, Diffusion Model, Image Generation, Human-in-the-loop

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1 INTRODUCTION

AIGC's breakthroughs in recent years have attracted widespread attention. Stable Diffusion [6], with DDPM [1] and DDIM [7] as the core technology, can efficiently generate beautiful images based on text descriptions, which has greatly changed the way people create. However, the controllability of the Stable Diffusion generating process is limited, and the image generated by text cannot meet the complex requirements, and there are defects in the details. In order to generate accurate images, Microsoft proposes a controllable

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image generation algorithm, Control-GPT [11], which uses GPT-4 [5] to generate TikZ code to create sketches, and guides Stable Diffusion to achieve a controllable text-to-image generation process. This method achieves coarse-grained control over the shape, size, and position of objects in the generated image, but fails to achieve fine-grained control.

The traditional method of computer-aided suspect portrait uses the pre-existing face shape, hairstyle, facial features and other face parts libraries in the system to make combination adjustments. This method improves the efficiency of portrait drawing, but is limited by the complexity and diversity of parts in the face parts libraries. It is difficult to cover all fine-grained personalized facial features, and only sketches or virtual face can be generated, unable to generate real-style face pictures.

In this paper, we propose CFTF, a controllable fine-grained face generation technology. It is completely based on the most cutting-edge cross-modal large model and Prompt Engineering [4, 8] technology. It can be used to achieve fine-grained control of facial details such as hair color, accessories, and pupil color by text. Based on CFTF, we construct an application scenario of suspect portraits, introduced the "Human-in-the-loop" collaboration mechanism, and introduced multiple rounds of interactions between the user and the face creation process to generate a profile that better reflects the user's detailed needs. Based on the above technologies and ideas, we have implemented a web application that provides a good user experience, and are committed to promoting our system to a wider range of application scenarios.

2 CFTF: CONTROLLABLE FINE-GRAINED TEXT2FACE

In this section, we will provide a detailed introduction to CFTF, including its sub-modules and some implementation details. Figure 2 shows the technical route of CFTF and three key modules:

2.1 Controller Module

As the controller of CFTF, controller module accepts the user's natural instructions and uses Prompt Engineering technology to help the large language model understand and interpret these instructions. It is responsible for assigning SD Prompt and SAM Prompt to Generation Module and Segment Module to realize indirect user control over them, driving the entire portrait drawing process.

2.2 Generation Module

This module can be regarded as the "painter" of the whole system, responsible for generating or optimizing face portraits. At its core, it is a Stable Diffusion variant with ControlNet [10] and LoRA [2] plugins to generate face images in a controlled manner based on normalized prompts. This module receives the SD Prompt from the Controller Module and the Mask output from the Segment Module, which contains information about the style and details of the portrait to be generated. Based on publicly available face recognition datasets and manually collected data, we constructed a dataset of about 10,000 face images and used BLIP [3] to describe facial features more finely, including hairstyle, eye color and more fine-grained features of the face. This dataset is used to fine-tune both ControlNet and LoRA.

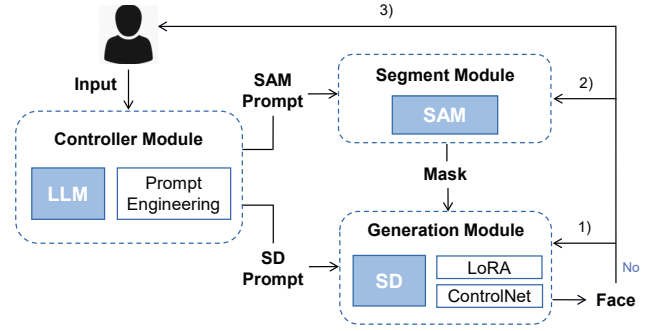


Figure 2: Diagram of CFTF's technical route and core modules. When the generated face is not satisfied, it will be used as the input of 1) Generation module, 2) Segment module, and 3) driving the user to input new fine-tuning text.

2.3 Segment Module

Segment module serves as an auxiliary component to the Generation Module, playing a crucial role in the process of facial repainting. When dissatisfied with the generated portrait, users can provide the Controller Module with a description of the desired refinements. The Controller Module then generates corresponding SAM Prompts for the Segment Module. The Segment Module analyzes the refinement requirements, segments the relevant areas, and generates masks required by the Generation Module. Consequently, the Generation Module selectively repaints specific regions, thereby achieving fine-grained control over facial generation.

3 SUSPECT PORTRAITS APPLICATION

Based on CFTF, we built a portrait generation application that can be used in police suspect portrait scenarios. Figure 1 provides an overview of the workflow. The basic process is as follows:

Step 1: Witnesses can draw a sketch (optional), and describe the suspect's facial features in words, such as hair color, skin color, accessories, etc.;

Step 2: The preliminary information (Sketch* + Text) provided by the user will be used as the original input of CFTF, and CFTF will generate multiple candidate face portraits;

Step 3: User can select the most desired one and provide further textual descriptions in the second round, focusing on details that need refinement, such as "wearing black-rimmed glasses."

In multiple rounds of human-computer interaction, we gradually output the expected suspect portraits based on CFTF's powerful controllable generation capabilities. This tightly coupled portrait drawing process fully demonstrates the "Human-in-the-loop" human-machine collaboration mechanism.

The full demo runs on a machine with an Nvidia GeForce RTX 3090 GPU. In order to meet the requirements of real-time generation, this application is accelerated with xformer [12] embedded in Stable Diffusion, and the generation of a single image takes about 3 seconds. We have provided an online demo experience link for live demonstrations, where users can interact with the system using both voice and text inputs for descriptive text. Sketches can be drawn by users on other devices and uploaded to the system.

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